Diffusion Cheat Sheet

1. Core Parameters & Schedules

Calculate these first before attempting heavier transitions.

- **Signal Retention:**

$$\alpha_t = 1 - \beta_t$$

- **Exam Keywords:** "Variance schedule", "noise schedule β_t", "retention at step t".

- **Cumulative Signal Retention:**

$$\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$$

- **Exam Keywords:** "Total signal", "jump from X_0", "arbitrary timestep t".

- **Signal-to-Noise Ratio (SNR)** ***(Added)***:

$$\text{SNR}_t = \frac{\bar{\alpha}_t}{1-\bar{\alpha}_t}$$

- **Exam Keywords:** "SNR at step t", "ratio of data to noise".

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2. Forward Process (Adding Noise)

- **The Iterative Step (Markov Chain):**

$$X_t = \sqrt{\alpha_t} X_{t-1} + \sqrt{1-\alpha_t}\epsilon_t$$

- **Distribution:**

$$q(x_t|x_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}x_{t-1}, \beta_t I)$$

- **Exam Keywords:** "Immediate next noisy state", "simulate one step forward", "given only previous state X_{t-1}".

- **The Direct Jump (Closed-Form):**

$$X_t = \sqrt{\bar{\alpha}_t} X_0 + \sqrt{1-\bar{\alpha}_t}\epsilon_t$$

- **Distribution:**

$$q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I)$$

- **Exam Keywords:** "Given clean image X_0", "distant timestep t", "relative to the beginning".
- **Isolating X_0 (Clean-Up):**

$$X_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}(X_t - \sqrt{1 - \bar{\alpha}_t}\epsilon)$$

- **Exam Keywords:** "Recover original clean image", "exact noise vector is known", "calculate predicted $\hat{X}_0$".

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3. Reverse Process (Training & Posterior)

- **True Posterior Variance:**

$$\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$$

- **Exam Keywords:** "True posterior variance", "variance conditioned on X_0", $\tilde{\beta}_t$.
- **True Posterior Mean (Given X_0 & X_t):**

$$\tilde{\mu}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}X_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}X_t$$

- **Exam Keywords:** "Expected value of previous state", "ground truth mean", "given original and noisy images".
- **True Posterior Mean (Given X_t & ϵ):**

$$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}}\left(X_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon\right)$$

- **Exam Keywords:** Same as above, but "given predicted noise ϵ_θ" instead of X_0.
- **DDPM Training Loss (Simplified):**

$$L_{simple}(\theta) = ||\epsilon - \epsilon_\theta||_2^2$$

- **Exam Keywords:** "Simplified objective", "MSE loss", "noise prediction error".
- **DDPM Training Loss (True ELBO):**

$$L_{vlb}(t) = \frac{\beta_t^2}{2\sigma_t^2\alpha_t(1-\bar{\alpha}_t)}||\epsilon - \epsilon_\theta(X_t, t)||_2^2$$

- **Exam Keywords:** "True bound", "ELBO loss", "pre-factor ignored by DDPM".

4. Sampling (DDPM vs. DDIM)

Standard DDPM (Markovian)

- **Denoising Step:**

$$X_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(X_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta \right) + \sigma_t z$$

- **Exam Keywords:** "Standard DDPM generation", "denoise exactly one step backward".
- **Final Step Boundary:** If $t = 1 \implies z = 0$
- **Exam Keywords:** "Final transition to X_0", "remove randomness at end".

DDIM (Non-Markovian Jumps)

- **Intermediate Clean Prediction:**

$$\hat{X}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}} (X_t - \sqrt{1-\bar{\alpha}_t} \epsilon_\theta)$$

- **Exam Keywords:** "Model's current prediction of clean image", "DDIM intermediate step".
- **DDIM Variance:**

$$\sigma_t = \eta \sqrt{\frac{1-\bar{\alpha}_{t'}}{1-\bar{\alpha}_t}} \sqrt{1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t'}}}$$

- **Exam Keywords:** "Stochasticity parameter η", "amount of randomness to inject".
- **DDIM Master Step-Skip:**

$$X_{t'} = \sqrt{\bar{\alpha}_{t'}} \hat{X}_0 + \sqrt{1 - \bar{\alpha}_{t'} - \sigma_t^2} \epsilon_\theta + \sigma_t z$$

- **Exam Keywords:** "Skip steps", "jump from t to t'", "deterministic sampling" (when $\sigma_t = 0$).
- **Final Step Boundary:** If $t' = 0 \implies \bar{\alpha}_{t'} = 1 \implies X_0 = \hat{X}_0$
- **Exam Keywords:** "DDIM jump lands at step 0", "final image output".

Improved DDPM

- **Variance Interpolation:**

$$\Sigma_\theta = \exp(v \log \beta_t + (1 - v) \log \tilde{\beta}_t)$$

- **Exam Keywords:** "Improved DDPM", "learnable variance", "interpolation factor v".

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5. Crucial Exam Caveats & Added Formulas

- **Boundary Conditions:**
- **Start:** X_0 is your pure data (unaltered image).
- **End:** As $T \to \infty$, $\bar{\alpha}_T \approx 0$. Therefore, $X_T \sim \mathcal{N}(0, I)$ (pure Gaussian noise).
- **DDPM Variance Selection (σ_t^2):** In standard DDPM papers, σ_t^2 is typically fixed to either β_t or $\tilde{\beta}_t$. If an exam asks for "untrained variance," assume one of these two.
- **The DDIM to DDPM Connection:** If you set $t' = t - 1$ (no skipping) and $\eta = 1$, the DDIM formula mathematically collapses perfectly into the standard DDPM step.
- **Deterministic DDIM:** If a question mentions "ODE solver," "deterministic generation," or $\eta = 0$, immediately set $\sigma_t = 0$. The z term disappears, making the process perfectly reversible.
- **The Reparameterization Trick:** If asked *why* we can write X_t in closed form, state that the sum of multiple independent Gaussians is also Gaussian. Variances add up, which collapses the iterative product into $\bar{\alpha}_t$.